



PRE-CALCULUS

ACADEMIC

Grade 11/12

Curriculum Guide

PRE-CALCULUS
Prerequisite: None

Course Description:

This Pre-Calculus course equips learners with essential mathematical knowledge and skills in preparation for Calculus courses, while fostering critical and analytical thinking, and problem-solving skills. Through exploration of graph transformations such as translations and reflections, functions including polynomials, rational, exponential, and logarithmic functions, and the analytic geometry of lines and circles, learners develop a deep understanding of mathematical principles and their real-world applications. Learning experiences include analyzing and solving complex problems, justifying solutions through logical reasoning, constructing proofs, and communicating mathematical ideas effectively using appropriate representations.

As preparation for higher education Calculus courses, learners will translate real-life situations into mathematical models using appropriate digital tools and technology. Equipped with 21st century skills, they apply their mathematical knowledge and understanding in preparation for future academic pursuits and professional careers in fields such as engineering, computer science, economics, statistics, finance, physics, and other sciences.

Elective: Academic

Time Allotment: 80 hours for a term

CONTENT DOMAIN	CONTENT STANDARDS <i>The learners demonstrate knowledge and understanding of ...</i>	LEARNING COMPETENCIES <i>The learners ...</i>
MEASUREMENT AND GEOMETRY (MG)	1. Lines and Circles	1. determine the equations of lines and circles from given geometric conditions (e.g., two points; a point and a slope; parallelism or perpendicularity; center and radius; and endpoints of a diameter). 2. represent real-life situations using equations of lines and circles. 3. analyze the geometric relationships between lines and circles using their equations.

CONTENT DOMAIN	CONTENT STANDARDS <i>The learners demonstrate knowledge and understanding of ...</i>	LEARNING COMPETENCIES <i>The learners ...</i>
		4. graph the solution sets of systems of equations involving lines and circles. 5. solve systems of equations involving lines and circles.
	2. Graph Transformations	6. apply translations and reflections to the graphs of equations 7. analyze the relationships between equations and their graphs by identifying patterns that affect orientation, symmetry, shape, and general behavior. 8. solve problems involving graphs of various types of equations and their transformations.
NUMBER AND ALGEBRA (NA)	3. Polynomial Functions	9. identify polynomial functions in third degree or higher. 10. apply the Remainder and Factor Theorems to determine the zeros of polynomial functions, where appropriate. 11. solve polynomial equations and inequalities. 12. graph and describe the characteristics of polynomial functions with and without graphing tools. 13. solve problems involving polynomial equations and functions.
Performance Standards The learners can investigate and model real-life situations using graphs of equations, and polynomial and rational functions. They can determine equations of lines and circles from geometric conditions, represent real-life situations using these equations, and analyze geometric relationships both algebraically and graphically. The learners can apply basic graph transformations, such as translations and reflections, to show and explain how changes in variables affect the movement, orientation, and overall behavior of graphs. They can also graph solution sets of systems and solve line-circle systems accurately. The learners can select appropriate function types for given contexts, define variables clearly, and state assumptions and constraints. They can represent these functions using equations, tables, and graphs, with appropriate use of technology, and analyze key features such as domain and range, intercepts, asymptotes, and points of intersection. The learners can solve polynomial equations using suitable algebraic methods such as Remainder and Factor Theorems, synthetic division and long division, and interpret these solutions in context. They can evaluate the reasonableness of results by checking against constraints, units, and graphs, and by comparing alternate approaches.		

CONTENT DOMAIN	CONTENT STANDARDS <i>The learners demonstrate knowledge and understanding of ...</i>	LEARNING COMPETENCIES <i>The learners ...</i>
The learners can communicate their findings through well-labeled graphs, precise notations, and clear explanations justifying methods and conclusions. In doing so, they demonstrate both theoretical understanding and practical applications of mathematics in context demonstrating readiness for higher-level studies and real-world challenges.		
NUMBER AND ALGEBRA (NA)	4. Rational Functions	14. differentiate rational expressions, equations, and functions. 15. graph rational functions and describe their intercepts, zeros, asymptotes, domain and range. 16. solve rational equations and real-life problems involving rational functions.
	5. Exponential and Logarithmic Functions	17. define the inverse of a function and enumerate its properties. 18. identify the inverse of simple one-to-one algebraic functions (linear, square root of linear, $y=x^3$, quotient of two linear functions). 19. sketch the graph of the inverse function given the graph of the original function. 20. define and illustrate exponential functions. 21. formulate the inverse of an exponential function using logarithms. 22. sketch graphs of exponential and logarithmic functions and identify their domains and ranges. 23. prove properties of logarithms. 24. solve exponential and logarithmic equations. 25. solve real-life problems using exponential and logarithmic functions and equations.

CONTENT DOMAIN	CONTENT STANDARDS <i>The learners demonstrate knowledge and understanding of ...</i>	LEARNING COMPETENCIES <i>The learners ...</i>
Performance Standards The learners can conduct an in-depth analysis of real-life situations involving rational, exponential, and logarithmic functions, such as resource optimization, electrical resistance, cooling rates, loan interests, and population growth. They can represent these relationships using appropriate models and provide justified solutions and interpretations. The learners can also define inverses of one-to-one functions, construct and interpret the inverse of an exponential function using logarithms and vice-versa, and sketch and analyze their graphs with correct domains and ranges. Furthermore, they can prove and apply the properties of logarithms to solve related equations and interpret key parameters in real-life contexts, including growth and decay, half-life, and doubling time. Using appropriate technology, they can investigate how changes in parameters can affect graph behavior and analyze connections between algebraic expressions and their graphical representations. Through these experiences, they can demonstrate the ability to connect abstract mathematical concepts to practical applications in areas such as growth and decay, signal processing, design, navigations, and optimization, while strengthening critical thinking, problem-solving, and readiness for advanced study and real-world challenges.		